

Observing resonance behavior of a temperature-regulated quartz crystal

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Introduction

Here we investigate the resonance of a temperature controlled quartz crystal to assess its use in timekeeping. The impedance of a crystal is known to drop dramatically at resonance, which we monitor electronically by connecting the crystal as the “other half” of a voltage divider, and sweep frequencies across the crystal while monitoring the voltage drop across it. After each sweep, we extract the centroid of the resonance, collecting 1000s after many days of computer-automated runs. We report a statistical analysis of the resonance position and observe a $Q \sim 10^5$, which can discipline a electronic oscillator to less than 1s/day.

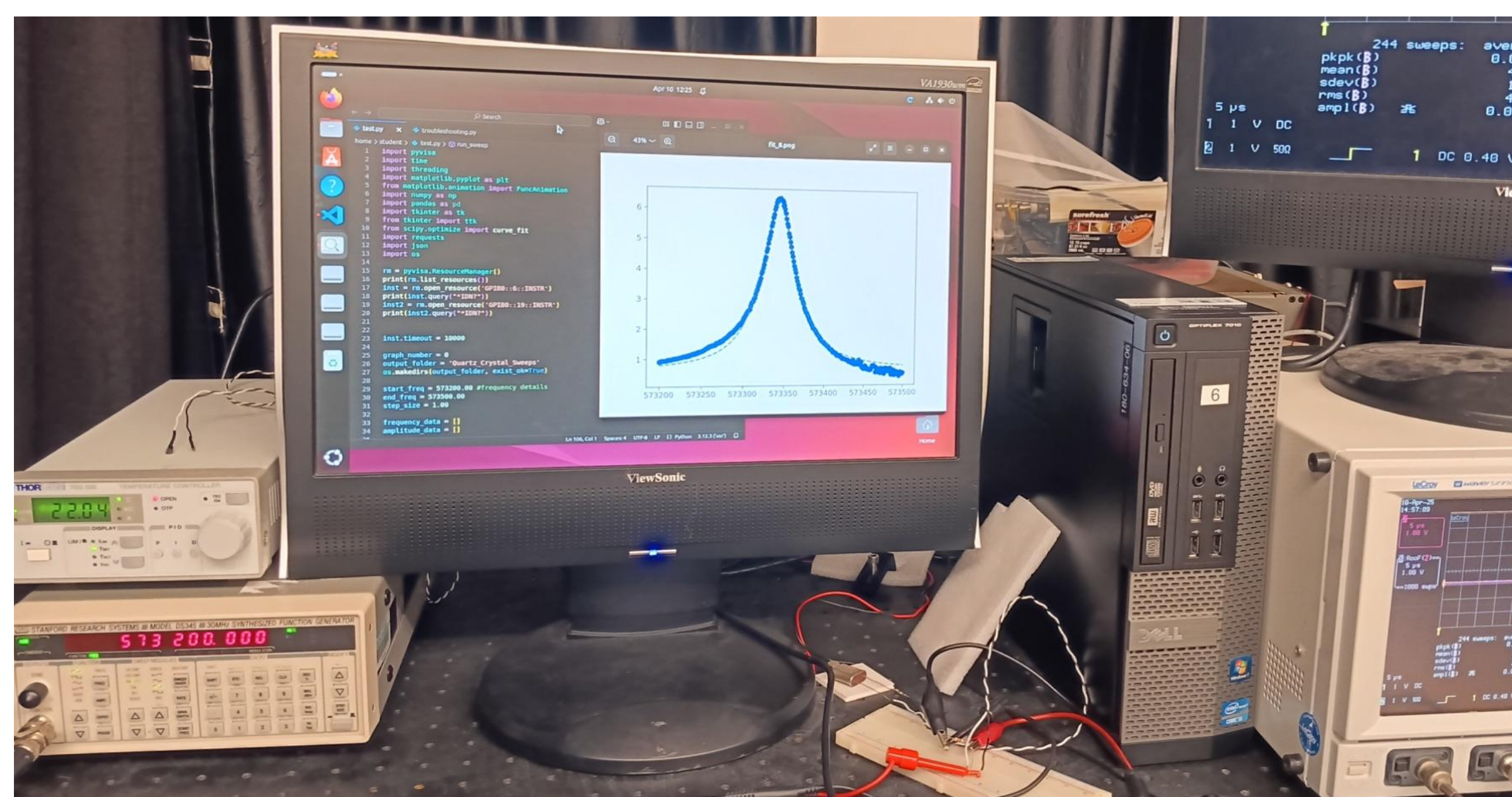
Methods

IN THE LAB:

- Varying frequencies are sent through a voltage-divider system to be read by the oscilloscope. The impedance of the crystal will drop at or near resonance frequency.
- A Python script was developed to automate the resonance sweeps from just below to just above resonance, while recording the voltage across the resistor.

ANALYSIS:

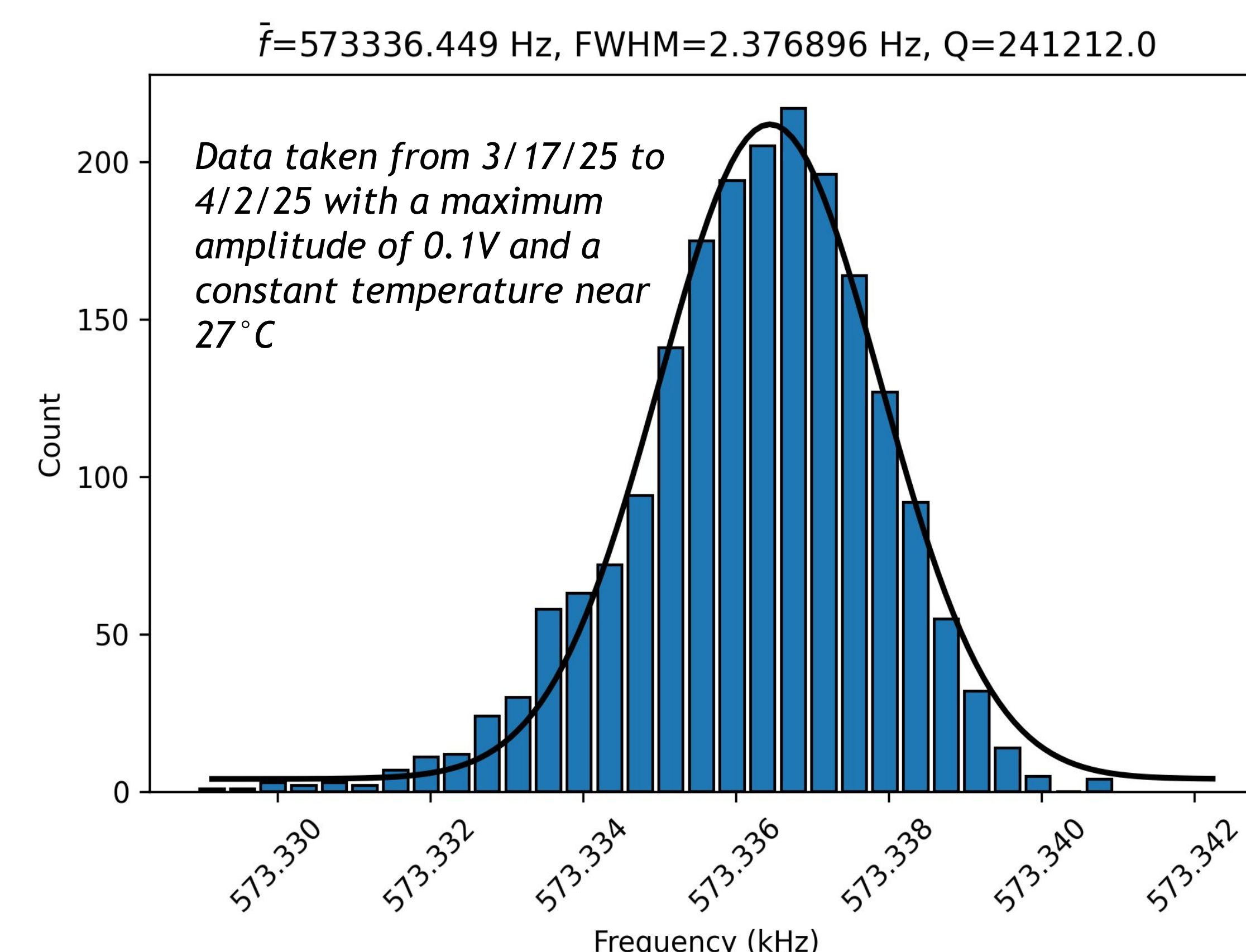
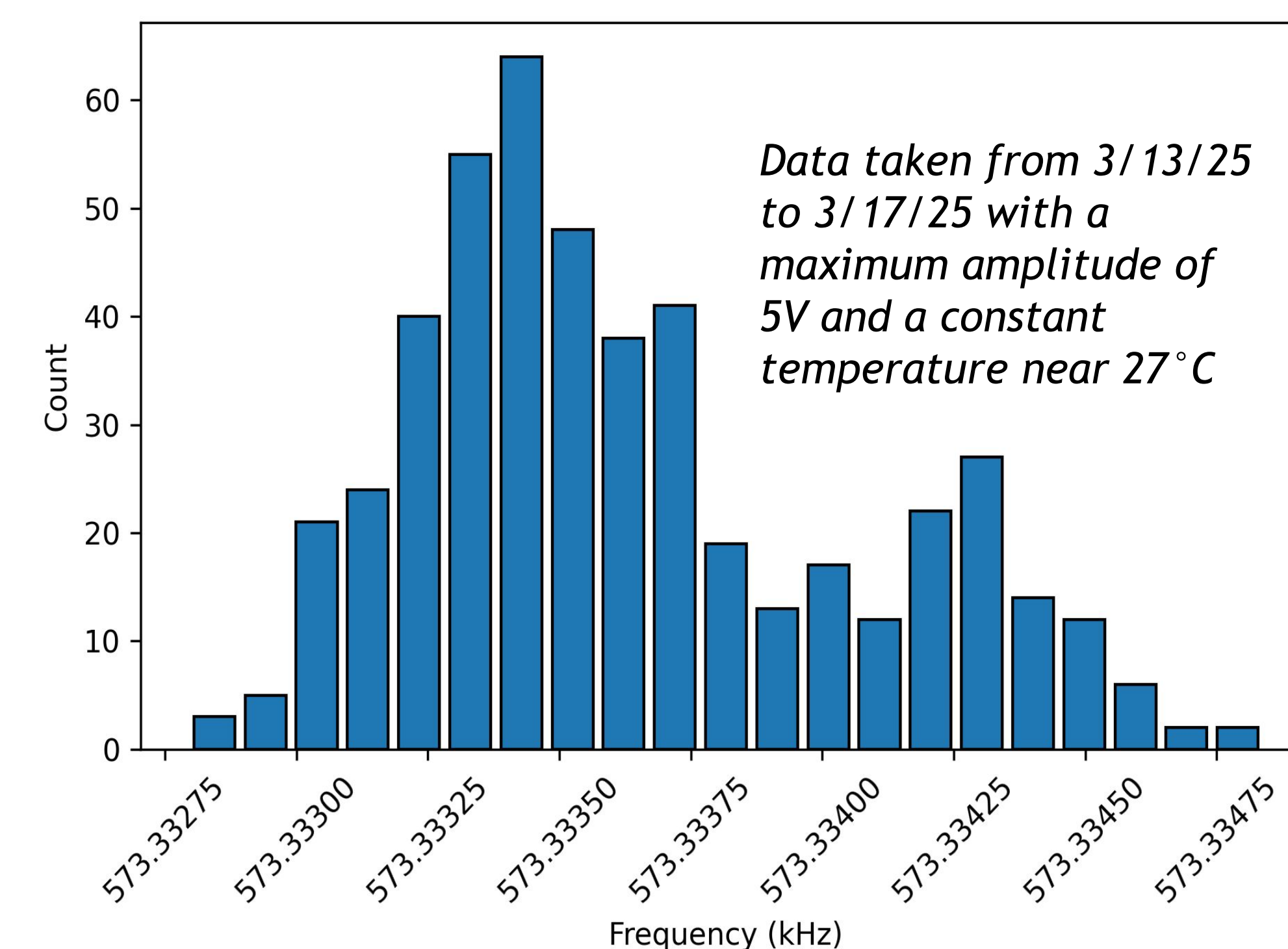
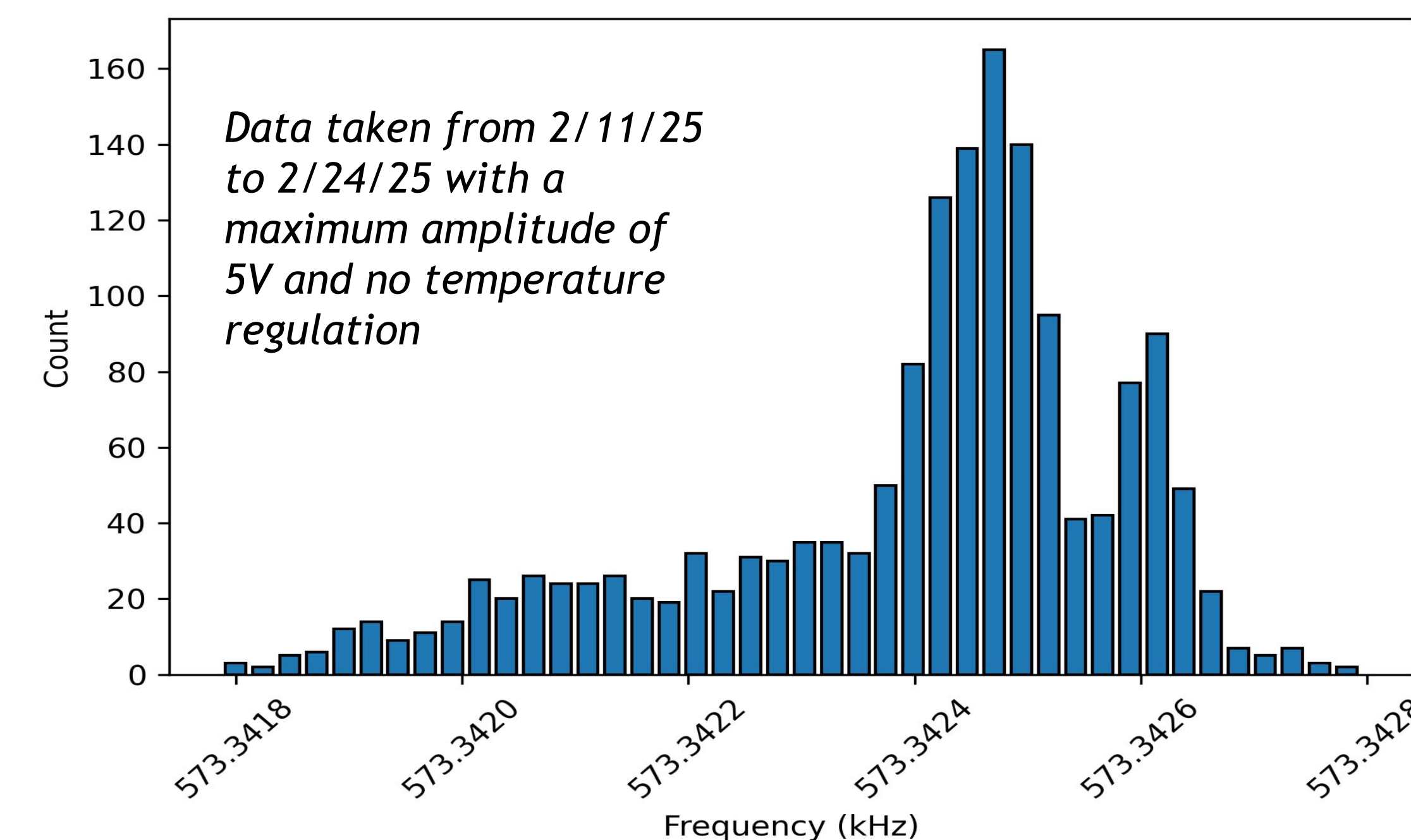
- Histogramming resonance centroids from 1000s of trials allows us to find a mean and standard deviation to determine quality factor or “Q”
- $Q = \Delta t / t$ and $Q = f_0 / \Delta f$
- $t = \Delta t / (f_0 / \Delta f)$



The equipment used to collect data for this project, along with the wiring.

Results

We worked on several laboratory configurations, with and without temperature control and with large and small amplitudes of the frequency sweep (top and middle plots). We seem to fall into the regime of random errors only when under temperature control with a small driving amplitude (bottom plot).



Conclusions / Discussions

Based on the data collected, the temperature of the quartz crystal was a significant factor in the precision the resonance position. We see a dramatic increase in precision of the resonance position with temperature control.

The driving amplitude of the frequency sweep caused a bimodal resonance structure, indicative of an over-driven system. A lower amplitude resulted in a single-mode resonance structure.

Over a continuous sampling for 15 days, we conclude that our crystal has a $Q = 2.5 \times 10^5$ at 27C, meaning if used in timekeeping applications, the crystal would allow for timekeeping to within 0.34 seconds/day.

Future Directions / Next Steps

We wish to compare findings of this work to atomic-disciplined oscillators. Also, we wish to investigate crystals with higher (i.e. MHz) resonance frequencies.

References

- **M. Lombardi** “The Accuracy and Stability of Quartz Watches,” Horological Journal, pp. 57-59, 2008.
- **Atwood, A. W. Jr.** (1933). *A Free Quartz Pendulum: Electrostatically Driven and Photoelectrically Controlled as a Means of Precision Timekeeping*. Master's thesis, California Institute of Technology. doi:10.7907/c5s0-qx08

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